Linear functions

Outline

Linear functions
- Definitions
- Gradient and level sets in the plane
- Graphs of linear functions

Linear functions

Introduction: Linear functions

- In this chapter we define **linear functions**, the functions that appear as objective functions in Linear Programming
- We compute their **gradient** and their **Hessian**, and we describe their **level sets**
- These objects tell us, geometrically, in which direction a linear function grows and where it is constant: this is the picture we will use for Linear Programming problems later in the course
- We conclude with examples in the plane: gradients, level sets and graphs

Definition 1: linear function

Given a (column) vector

$$\underbrace{\begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}}_{\boldsymbol{c}} \in \mathbb{R}^n$$

the associated (scalar) function

$$f : \mathbb{R}^n \to \mathbb{R}, \quad f : \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{\boldsymbol{x}} \mapsto \underbrace{\sum_{j=1}^{n} c_j x_j}_{\boldsymbol{c'x}}$$

is called a ***linear function****.*

From the function to its derivatives

- A linear function is completely described by its vector of coefficients $\boldsymbol{c}$
- We now look at its first and second partial derivatives: the **gradient** and the **Hessian**
- We will see that both are very simple: they do not depend on the point $\boldsymbol{x}$

Gradient

Definition 2: gradient of a linear function

*The **gradient** (vector of the partial derivatives) of the linear function $f(\boldsymbol{x}) = \boldsymbol{c}'\boldsymbol{x}$ is:*

$$\nabla f(\boldsymbol{x}) = \begin{pmatrix} \frac{\partial f(\boldsymbol{x})}{\partial x_1} \\ \frac{\partial f(\boldsymbol{x})}{\partial x_2} \\ \vdots \\ \frac{\partial f(\boldsymbol{x})}{\partial x_n} \end{pmatrix} = \underbrace{\begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}}_{\boldsymbol{c}}$$

Hessian (1/2)

Definition 3: Hessian of a linear function

*The **Hessian** (matrix of the second partial derivatives) of the linear function $f(\boldsymbol{x}) = \boldsymbol{c}'\boldsymbol{x}$ is the $n \times n$ matrix:*

$$\nabla^2 f(\boldsymbol{x}) = \begin{pmatrix} \frac{\partial^2 f(\boldsymbol{x})}{\partial x_1 \partial x_1} & \frac{\partial^2 f(\boldsymbol{x})}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f(\boldsymbol{x})}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f(\boldsymbol{x})}{\partial x_2 \partial x_1} & \frac{\partial^2 f(\boldsymbol{x})}{\partial x_2 \partial x_2} & \cdots & \frac{\partial^2 f(\boldsymbol{x})}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f(\boldsymbol{x})}{\partial x_n \partial x_1} & \frac{\partial^2 f(\boldsymbol{x})}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f(\boldsymbol{x})}{\partial x_n \partial x_n} \end{pmatrix}$$

Hessian (2/2)

Definition: Hessian of a linear function (continued)

For a linear function all these entries are zero:

$$\nabla^2 f(\boldsymbol{x}) = \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}$$

since every first partial derivative $\frac{\partial f(\boldsymbol{x})}{\partial x_j} = c_j$ *is a constant.*

Where the function is constant

- The gradient tells us in which direction the function changes
- We now look at the opposite question: the sets of points where the function does **not** change

Level sets

Definition 4: level sets of a linear function

*The **level sets** of the linear function $f(\boldsymbol{x}) = \boldsymbol{c}'\boldsymbol{x}$ (the sets of points in which the function is **constant**) are:*

$$\left\{ \boldsymbol{x} \in \mathbb{R}^n : \quad \boldsymbol{c}' \boldsymbol{x} = k \right\}, \qquad \forall k \in \mathbb{R}$$

Gradient, Hessian and level sets

- The gradient of a linear function is the same vector $\boldsymbol{c}$ at every point and, accordingly, its **Hessian** (matrix of the second partial derivatives) is the zero matrix everywhere.
- The level sets of a linear function are the **hyperplanes** perpendicular to the gradient.

Gradient and level sets in the plane

- We now see these definitions at work for $n = 2$, where everything can be drawn
- Two examples, with opposite vectors of coefficients: $\boldsymbol{c} = \begin{pmatrix} 8 \\ 4 \end{pmatrix}$ and $\boldsymbol{c} = \begin{pmatrix} -8 \\ -4 \end{pmatrix}$
- In each example: the function, its gradient, and a figure with the gradient and two level sets

Example (1/3)

Gradient and level sets of linear functions

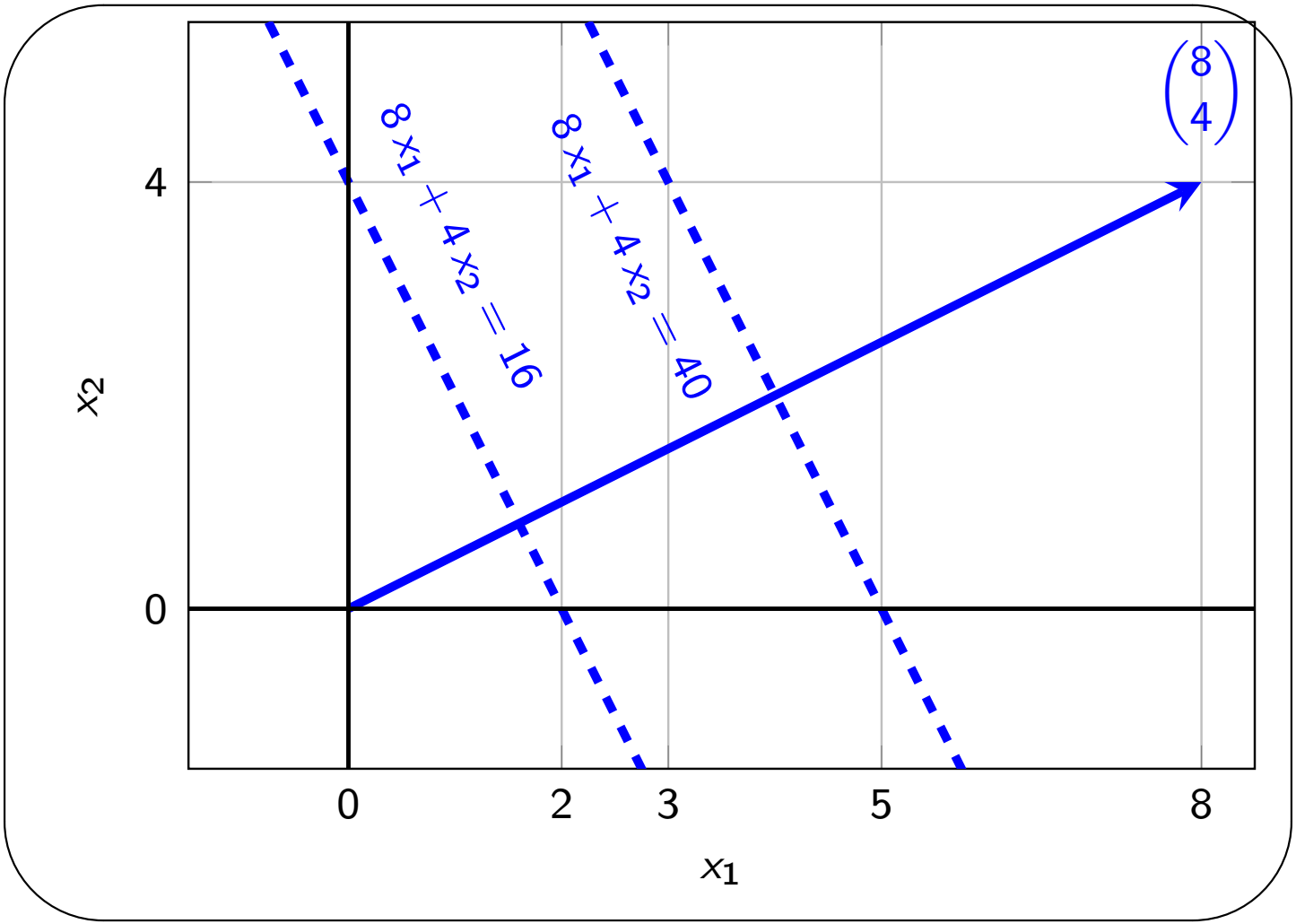
With $\boldsymbol{c} = \begin{pmatrix} 8 \\ 4 \end{pmatrix}$, the associated linear function is:

$$f : \mathbb{R}^2 \to \mathbb{R}, \quad f : \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mapsto \begin{pmatrix} 8 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 8\, x_1 + 4\, x_2$$

Gradient:

$$\nabla f \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right) = \begin{pmatrix} 8 \\ 4 \end{pmatrix}$$

Example (2/3)


$8x_1 + 4x_2 = 16$
$8x_1 + 4x_2 = 40$
$\begin{pmatrix} 8 \\ 4 \end{pmatrix}$
4
0
x_2
0
2
3
5
8
x_1

Example (3/3)

The figure shows:

- The gradient evaluated at $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$: $\nabla f\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 8 \\ 4 \end{pmatrix}$ (blue arrow)
- Two level sets (blue dashed lines): $8x_1 + 4x_2 = 16$ and $8x_1 + 4x_2 = 40$
- The level sets are lines (hyperplanes of $\mathbb{R}^2$) **perpendicular** to the gradient
- Moving in the direction of the gradient, we meet level sets with larger values of k: first 16, then 40

Opposite coefficients

We now change the sign of every coefficient:

$$\boldsymbol{c} = \begin{pmatrix} -8 \\ -4 \end{pmatrix}.$$

The gradient points in the **opposite** direction,
and the level sets are still perpendicular to it.

Example (1/3)

Gradient and level sets of linear functions

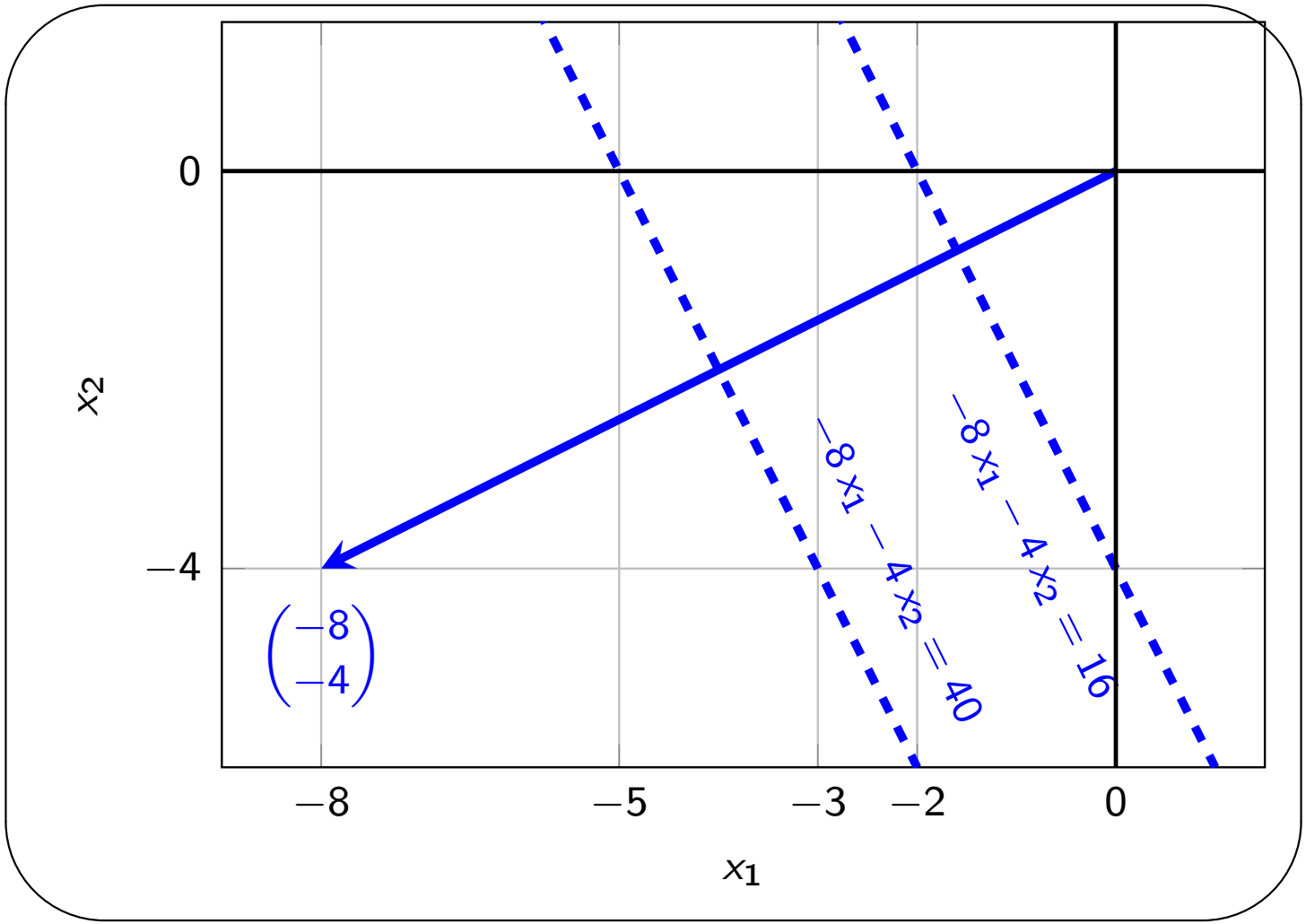
With $\boldsymbol{c} = \begin{pmatrix} -8 \\ -4 \end{pmatrix}$, the associated linear function is:

$$f : \mathbb{R}^2 \to \mathbb{R}, \quad f : \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mapsto \begin{pmatrix} -8 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = -8\, x_1 - 4\, x_2$$

Gradient:

$$\nabla f \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right) = \begin{pmatrix} -8 \\ -4 \end{pmatrix}$$

Example (2/3)


0
−4
x2
−8
−5
−3
−2
0
x1
$\begin{pmatrix} -8 \\ -4 \end{pmatrix}$
$-8x_1 - 4x_2 = 40$
$-8x_1 - 4x_2 = 16$

Example (3/3)

The figure shows:

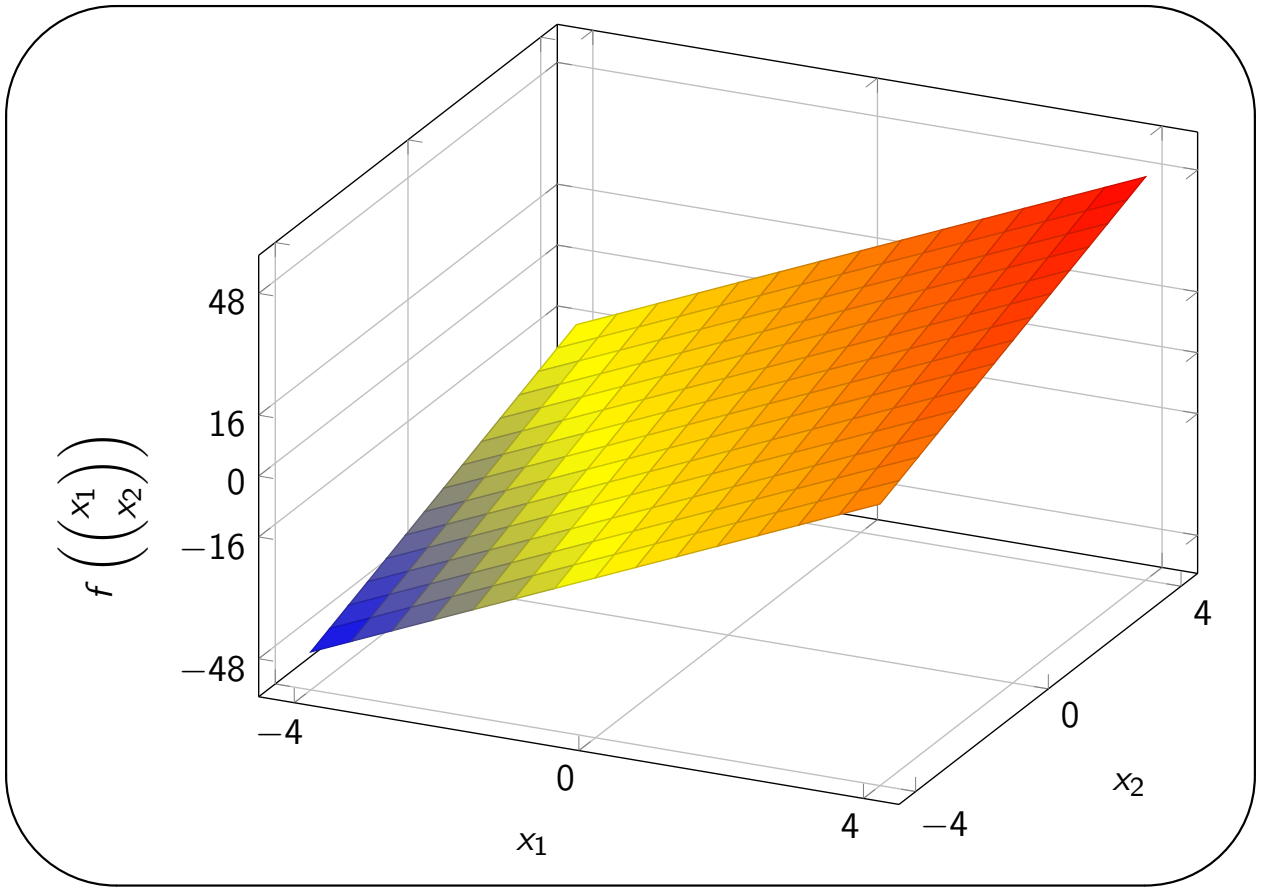
- The gradient evaluated at $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$: $\nabla f\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} -8 \\ -4 \end{pmatrix}$ (blue arrow)
- Two level sets (blue dashed lines): $-8x_1 - 4x_2 = 16$ and $-8x_1 - 4x_2 = 40$
- Again the level sets are lines **perpendicular** to the gradient
- Now the gradient points towards the lower left: the values 16 and 40 are reached by moving in that direction

Graphs of linear functions

- For $n = 2$ we can also draw the **graph** of a linear function: the points $\left(x_1, x_2, f\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right)\right)$ in three dimensions
- The graph of a linear function of two variables is a **plane** through the origin
- We draw it for the same two vectors $\boldsymbol{c} = \begin{pmatrix} 8 \\ 4 \end{pmatrix}$ and $\boldsymbol{c} = \begin{pmatrix} -8 \\ -4 \end{pmatrix}$

Example

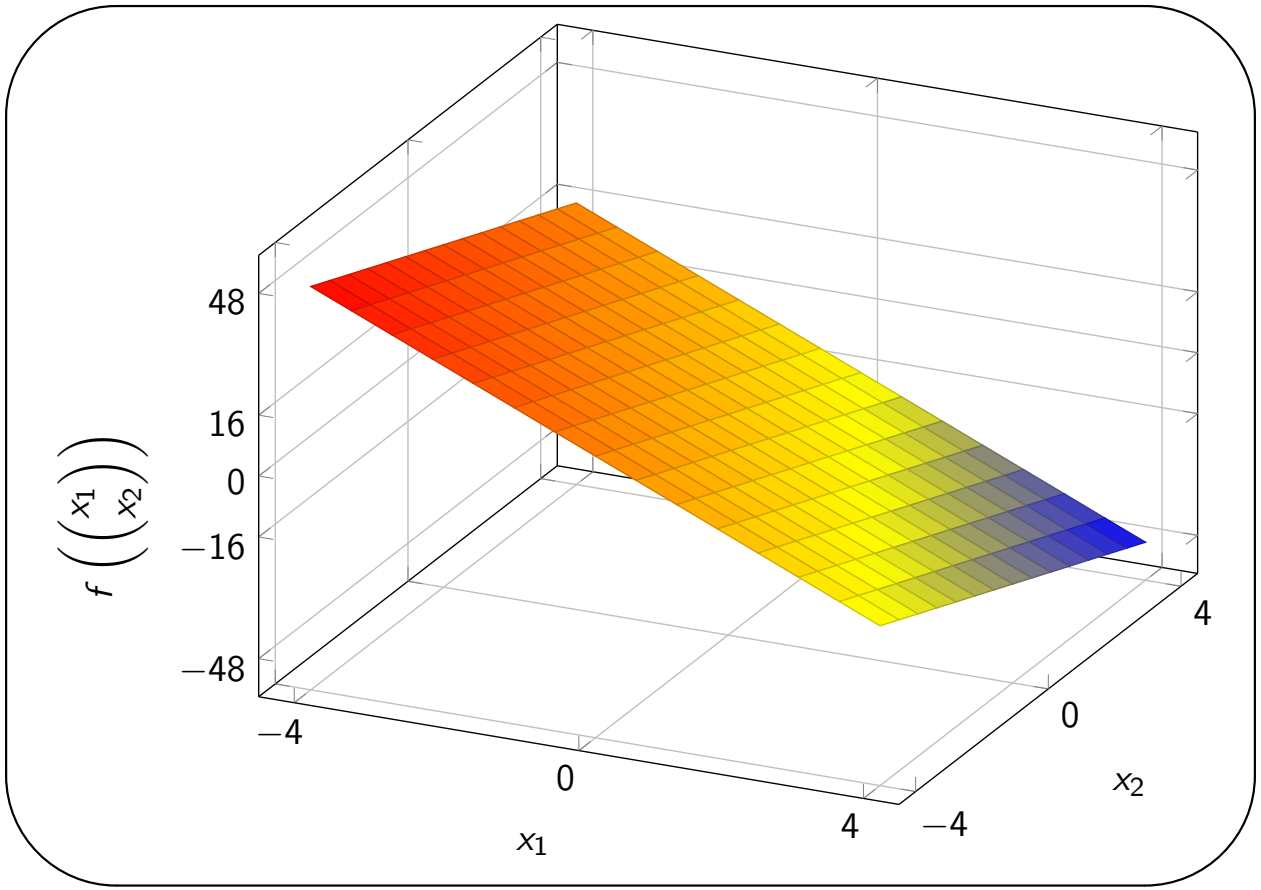
Graphs of linear functions. With $\boldsymbol{c} = \begin{pmatrix} 8 \\ 4 \end{pmatrix}$, we have $f\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = 8\, x_1 + 4\, x_2$. Its graph is:



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Example

Graphs of linear functions. With $\boldsymbol{c} = \begin{pmatrix} -8 \\ -4 \end{pmatrix}$, we have $f\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = -8\, x_1 - 4\, x_2$. Its graph is:



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Summary (1/2)

Linear function:

- Given $\boldsymbol{c} \in \mathbb{R}^n$, $f(\boldsymbol{x}) = \boldsymbol{c}'\boldsymbol{x} = \sum_{j=1}^{n} c_j x_j$

Gradient and Hessian:

- $\nabla f(\boldsymbol{x}) = \boldsymbol{c}$ at every point
- $\nabla^2 f(\boldsymbol{x})$ is the zero matrix everywhere

Level sets:

- $\{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{c}'\boldsymbol{x} = k\}$, for every $k \in \mathbb{R}$
- They are the **hyperplanes** perpendicular to the gradient

Summary (2/2)

In the plane ($n = 2$):

- The gradient is an arrow, the level sets are parallel lines perpendicular to it
- With $\boldsymbol{c} = \begin{pmatrix} 8 \\ 4 \end{pmatrix}$ and $\boldsymbol{c} = \begin{pmatrix} -8 \\ -4 \end{pmatrix}$ the gradients point in opposite directions

Graphs:

- The graph of a linear function of two variables is a plane in three dimensions